

1 Natural Units

“Natural units” are based on universal and fundamental constants of nature and make our equations look simpler and more elegant. This approach treats dimensionful constants (are they even constant if they depend on your choice of units?) as conversion factors between different physical quantities, leaving us with typically a single or a few fundamental units to keep track of (typically energy, like electronvolts, eV).

As you learned in your Special Relativity course, the speed of light in vacuum,

$$c = 299\,792\,458 \text{ m/s} \simeq 1 \text{ foot/ns}, \quad (1)$$

is a constant for all observers; therefore, it is as universal a quantity as we could hope for. In natural units, we enforce c to be unity, such that

$$c \stackrel{!}{=} 1 \implies 1 \text{ s} = 299\,792\,458 \text{ m}. \quad (2)$$

That is, by setting $c = 1$, we have simply stated that meters and seconds are equivalent measures of something (call it a space-time distance, for instance), and that it takes a lot of meters to have a comparable “space-time” second. By imposing Eq. (2), we have effectively reduced the number of fundamental units we need to keep track of by one: you may now use only meters or only seconds to fully specify a physical quantity.



As a consequence, speed or velocity are dimensionless quantities since their magnitudes are always interpreted as a fraction of the speed of light, which is just unity ($c = 1$).

The relativistic energy for a particle with mass m and momentum p is

$$E^2 = m^2 c^4 + p^2 c^2. \quad (3)$$

In natural units,

$$E^2 = m^2 + p^2. \quad (4)$$

The equivalence between mass, momentum, and energy becomes much clearer in this way: E , m , and p can all be expressed in terms of energy units.

Another universal constant we can leverage to simplify our units is Planck’s constant h (in particular, its reduced form will make less of a mess):

$$\hbar = \frac{h}{2\pi} = 1.054571817 \times 10^{-34} \text{ J.s} = 6.582119569 \times 10^{-16} \text{ eV/Hz}, \quad (5)$$

where we used the definition of an electronvolt in the second equality ($1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}$ is the energy acquired by an electron in an electric potential of one volt. Hence, electronvolt is a unit of energy.) By setting

$$\hbar \stackrel{!}{=} 1 \implies 1 \text{ Hz} = \text{s}^{-1} = 6.582119569 \times 10^{-16} \text{ eV}. \quad (6)$$

we make it clear that $\hbar$ relates energy to frequency (or inverse time) in a very fundamental way. This follows the physics since the energy of a photon is given by

$$E = h\nu = \hbar\omega = \frac{2\pi\hbar}{T} = \frac{2\pi\hbar c}{\lambda}, \quad (7)$$

where ν is the frequency of the photon, $\omega = 2\pi\nu$ is its angular frequency, T is its period, and λ is its wavelength. In fact, from all equalities above, you can deduce that $\hbar = 1$ and $c = 1$ put energy, frequency, time, and distance all on the same footing.

We started with three units: energy (eV), time (s), and distance (m). By setting $\hbar = 1$ and $c = 1$, we imposed two independent constraints on the system, leaving just one independent fundamental unit. We will pick it to be energy.



By convention, when working with natural units, we typically express all quantities in electronvolt (eV) or powers thereof. When necessary, we convert back to meters, seconds, or whatever else, by making use of the definition of $\hbar$ in Eq. (5), c in Eq. (2), or other convenient factors such as:

$$\hbar c \simeq 197 \text{ MeV}\cdot\text{fm} = 0.197 \text{ GeV}\cdot\text{fm} = 1.97 \times 10^{-5} \text{ eV}\cdot\text{cm}. \quad (8)$$



The reduced Compton wavelength of a massive particle is

$$\frac{\lambda}{2\pi} = \frac{\hbar}{m_e c}. \quad (9)$$

What is $\lambda/2\pi$ for electrons, $m_e = 0.511 \text{ MeV}$? How about for a proton, $m_p = 938 \text{ MeV}$? Express your answers in femtometers ($1 \text{ fm} = 10^{-15} \text{ m}$).



The mass of a proton is approximately $m_p \simeq 938 \text{ MeV}$. Using natural units, find the mass of a proton at rest in grams. How many protons in a gram?



The Hubble constant is $H_0 \simeq 75 \text{ (km/s)/Mpc}$. What distance does H_0^{-1} correspond to in Mpc? This is approximately the size of our causal horizon. What is H_0 in eV?

The above is the bulk of what we will need for our calculations, but there are two more examples that are worth mentioning. In Maxwell's equations, we also have the permittivity of free space ϵ_0 :

$$\epsilon_0 = 8.8541878128 \times 10^{-10} \text{ C}^2/(\text{N} \cdot \text{m}^2) \simeq 1.418 \times 10^{-32} \text{ C}^2/(\text{eV} \cdot \text{cm}). \quad (10)$$

This too, we will set to unity, $\epsilon_0 = 1$. In this case, the new constraint relates, for example, Coulombs (C), electronvolts (eV), and centimeters (cm). From $c = \hbar = 1$ and Eq. (8), we already know that centimeters and electronvolts are inversely related such that their product comes out dimensionless. Therefore, Setting $\epsilon_0 = 1$ has trivialized Coulombs as a unit and has set it to a dimensionless value of $\text{C} \simeq 5.28 \times 10^{-19}$.



In natural units, electric charge, usually expressed in terms of the elementary charge $e = 1.602176634 \times 10^{-19} \text{ C}$, is dimensionless. From Eq. (10), you can show that

$$e \simeq 0.303 \implies \alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137}. \quad (11)$$

Now, that's a constant! (Or is it?)

Let us connect this permittivity discussion to a familiar problem. Consider the energy ΔE gained by an electron of elementary charge $q_1 = e$ in an electric potential V sourced by a point

charge $q_2 = Qe$ at distance r away. In natural units, it is simply given by

$$\Delta E = q_1 \cdot V = \frac{q_1 q_2}{4\pi r} = \frac{\alpha Q}{r}. \quad (12)$$

This should feel more natural, the energy gained depends purely on the distance proportionally to dimensionless quantities: the charges and the fine-structure constant $\alpha \simeq \frac{1}{137}$. The latter appears here to characterize the strength of the electromagnetic interaction.

A final example is Boltzmann's constant k_B :

$$k_B = 1.380649 \times 10^{-23} \text{ J/K} = 8.617333262 \times 10^{-5} \text{ eV/K}. \quad (13)$$

Recall that the typical kinetic energy of a particle in an ideal gas of temperature T is $E \sim 3k_B T/2$. By setting $k_B = 1$, we find that the temperature is simply a measure of energy.



What is the typical energy of a photon in the Cosmic Microwave Background given that its temperature is $T \simeq 2.7 \text{ K}$? Express your answer in eV. How does this compare to the typical energy of a photon in visible light, which has a wavelength of about 500 nm? And to the ground state energy of the hydrogen atom?



The typical energy of a photon in the Cosmic Microwave Background (CMB) can be estimated using the relation $E \sim k_B T$. Given that $T \simeq 2.7 \text{ K}$, we find

$$E \sim k_B T \simeq 8.617333262 \times 10^{-5} \text{ eV/K} \times 2.7 \text{ K} \simeq 2.33 \times 10^{-4} \text{ eV}. \quad (14)$$

In comparison, the energy of a photon in visible light with a wavelength of about 500 nm is given by

$$E \sim \frac{hc}{\lambda} \simeq \frac{(4.135667696 \times 10^{-15} \text{ eV s})(3 \times 10^{10} \text{ cm/s})}{500 \times 10^{-7} \text{ cm}} \simeq 2.48 \text{ eV}. \quad (15)$$

Finally, the ground state energy of the hydrogen atom is approximately $E_1 \simeq -13.6 \text{ eV}$.